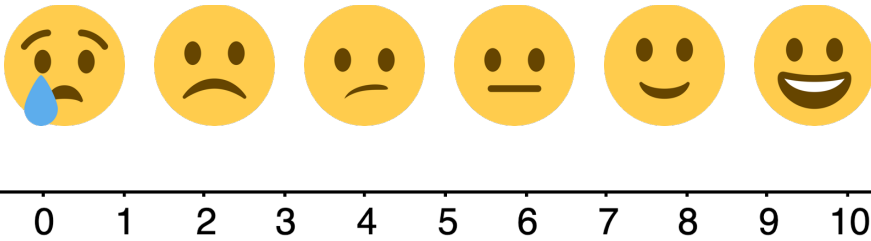




6.7.6 Wrap-Up: Reflection

1. Use the emoji scale to describe how confident you feel in writing a two-variable inequality in different forms when given the graph of a solution set.



2. Explain your selection.

Unit 6, Lesson 8: Writing a Two-Variable Linear Inequality to Represent a Real-World Context



Benchmarks

MA.912.AR.2.7 Write two-variable linear inequalities to represent relationships between quantities from a graph or a written description within a mathematical or real-world context.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.



Learning Targets

- ☐ I can write a linear inequality given a real-world context.
- ☐ I can interpret solutions to problems as viable or non-viable options.



6.8.1 Warm-Up: Bell Work

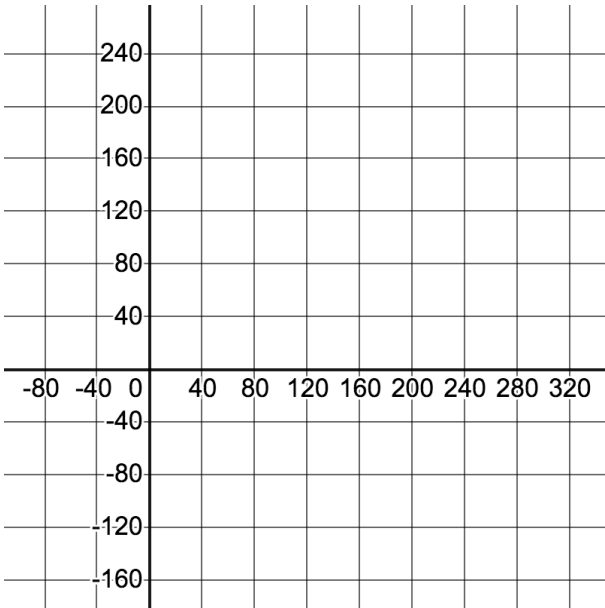
In the 2021 cookie-selling season, all the Girl Scout councils in Florida priced their cookies as shown in the table.



Core Varieties		Specialty Cookies	
Samoas, Thin Mints, Do-si-dos, Tagalongs, Trefoils, and Lemon-Ups	\$5 each	Toffee-tastic and S'mores	\$6 each

1. One troop in the Girl Scouts of Tropical Florida council makes a goal to sell at least \$1,200 in cookies. They write the inequality $5x + 6y \geq 1,200$ to determine how many boxes they need to sell, where x is the number of boxes of the core varieties and y is the number of boxes of the specialty cookies they sell.

- a. Graph the solution set of the inequality.



- b. Explain why the mathematical solution $(212, 83.5)$ is not viable in the context of selling cookies.



6.8.2 Collaboration: Building a Garden

1. Alissa wants to build a rectangular garden box in her yard. She needs the box to have three separated compartments to plant tomatoes, beets, and okra. The compartments do not need to be the same size. She has 48 feet (ft) of wood planking to build the garden box.



Alissa draws two possible designs shown in the table. Each square represents 1 ft^2 .

- a. Explain whether Alissa would have enough planking to make the garden box with these designs.

Design	Does she have enough planking?

- b. Work with your partner to sketch two designs that Alissa could use that meet her constraints. For each design, give the amount of wood she would use and explain or show how you got that amount.

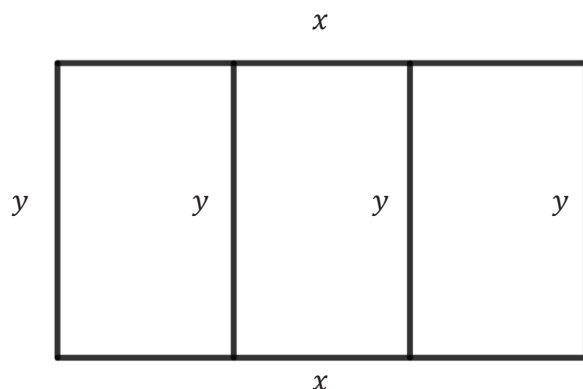
Design	How much planking does she need?



6.8.3 Guided Instruction: Writing Linear Inequalities in Two Variables to Represent Real-World Contexts

1. Alissa's garden scenario can be modeled by a two-variable linear inequality.
 - a. Describe what the two variables would represent in the context of Alissa's garden.

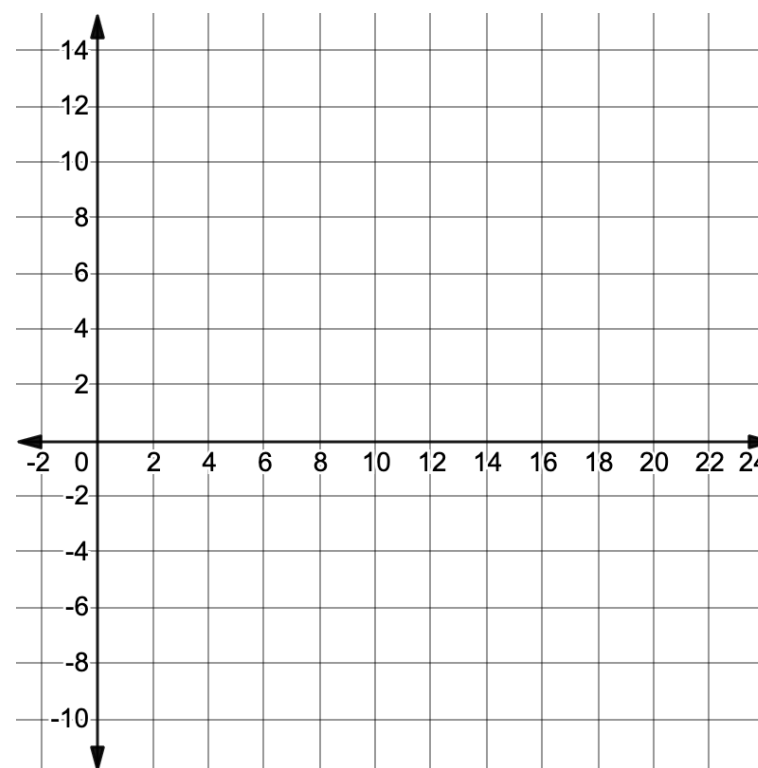
- b. The diagram below shows a possible model of Alissa's garden with sides labeled as x and y .



Write an expression in terms of x and y that represents the total amount of wood planking Alissa needs to construct the garden box.

- c. Recall Alissa had 48 ft of planking available to use. Write a two-variable inequality that Alissa could use to find possible dimensions for her garden.
 - d. Explain how you chose which inequality symbol to use.

- e. Graph the inequality.



- f. The three ordered pairs given in the table are solutions to the mathematical inequality. For each one, explain whether or not it is a viable solution in the real-world context.

Ordered Pair	Is it viable in the real-world context?	Explanation
$(-4, 7)$	☐ Yes ☐ No	
$(4\frac{1}{2}, 6)$	☐ Yes ☐ No	
$(14.5, 0)$	☐ Yes ☐ No	

- g. Explain what a solution on the boundary line of the inequality would mean in the real-world context.



6.8.4 Guided Practice: Writing Linear Inequalities in Two Variables to Represent Real-World Contexts

- The Miami Marlins baseball team moved into Marlins Park in 2012. That season, the average ticket price at Marlins Park was \$29.62. It costs approximately \$300,000 to operate the stadium for each game.
 - Write an expression to represent the amount of money made in ticket sales for a game, where x is the number of tickets sold.
 - Write an expression to represent the Marlins' profit from one game if tickets sales and operating costs were the only factors.
 - Write an inequality that relates x and y , where y is the profit from one game if ticket sales and operating costs were the only factors.

Note that the profit from a game is actually greater than the expression you wrote in Part B because there are other sources of revenue at a baseball game, such as concessions and merchandise.

d. Explain what the solution $(1,000, -250)$ would represent in the Marlins' situation.

e. Explain whether either of the following would be a viable solution in the Marlins' context.

$(1,255.85, 12,000)$ or $(10,814, 20310.68)$

f. For a solution to be viable in this context, which of the following statements must be true?

- ☐ $x \leq 0$
- ☐ $x \geq 0$
- ☐ $y \leq 0$
- ☐ $y \geq 0$

Explain your choice.

g. Describe any additional constraints on x and y in the real-world context.



6.8.5 Try It: Qualifying for a Loan

Banks consider a number of factors when determining whether to grant someone a personal loan. Some of these factors include credit score, income or employment history, debt-to-credit ratio, and liquid assets (possessions or investments that could easily be converted to cash).

1. Suppose a bank requires a minimum combined balance of \$800 in your checking and savings accounts as one of the liquid assets to be considered for the loan.
 - a. Determine whether each person meets the liquid assets minimum to be considered for a loan.

Account Information	Combined Balance	Meets liquid assets minimum?
Angelina has \$372.25 in her checking account and \$427.75 in her savings account.		☐ Yes ☐ No
Bashar overdraw his checking account, which now has a balance of $-\$250$. He has \$1,172 in his savings account.		☐ Yes ☐ No
Jayla has \$732.55 in her checking account, but doesn't have a savings account.		☐ Yes ☐ No

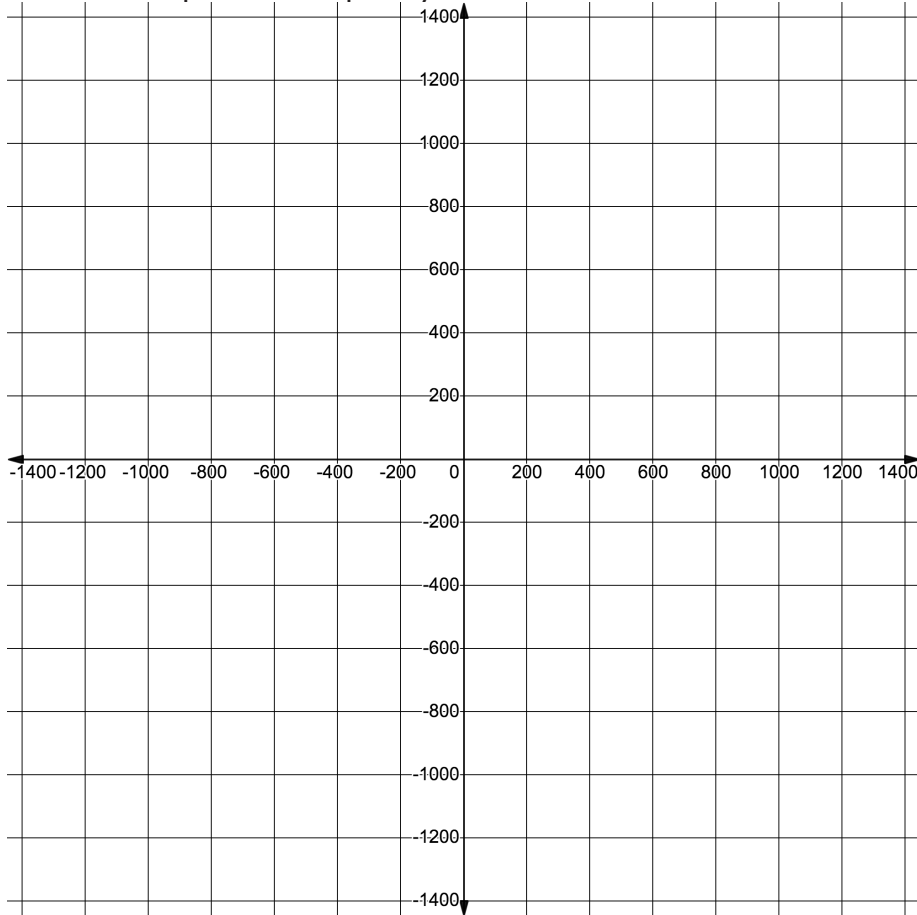
- b. Define two variables for this situation.

Let $x =$ _____.

Let $y =$ _____.

- c. Write an inequality in terms of x and y that could be used to determine whether the person meets the minimum liquid assets requirement for the loan.

- d. Graph the inequality.



- e. Complete the statement.

To be a viable solution in this context, the x and y values must both be rounded to ____ decimal places.



6.8.6 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found multiple solutions	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐